IN-CLASS QUIZ

AVL Trees

Balance and Insertion

4 multiple-choice questions · about 1 minute each

BEFORE WE START

Scan to answer

Open the form on your phone and keep it open for all four questions.

https://forms.gle/awUdzbW4Tj27YibP8

QUESTION 1

≈ 1 minute

Using the convention that a single node has height 0, what is the maximum possible height of an AVL tree containing exactly 7 nodes?

A 2

B 3

C 4

D 6

QUESTION 2

≈ 1 minute

Starting from an empty AVL tree, you insert 30, then 10, then 20. What restores the AVL property, and what is the resulting root?

A. Single rotation: right rotation at 30; root becomes 10

B. Single rotation: left rotation at 10; root remains 30

C. Double rotation: left rotation at 10, followed by right rotation at 30; root becomes 20

D. Double rotation: right rotation at 30, followed by left rotation at 10; root becomes 30

E. No rotation is needed; the tree is already balanced

QUESTION 3

≈ 1 minute

Insert the keys 10, 20, 30, 40, 50, 25 in that order into an initially empty AVL tree. What key is at the root at the end?

A 20

B 25

C 30

D 40

QUESTION 4

≈ 1 minute

A single key is inserted into an AVL tree of n nodes. Counting a double rotation as two single rotations, what is the maximum number of rotations needed to restore the AVL property?

A 1

B 2

C About $\log_2 n$

D About $2 \log_2 n$